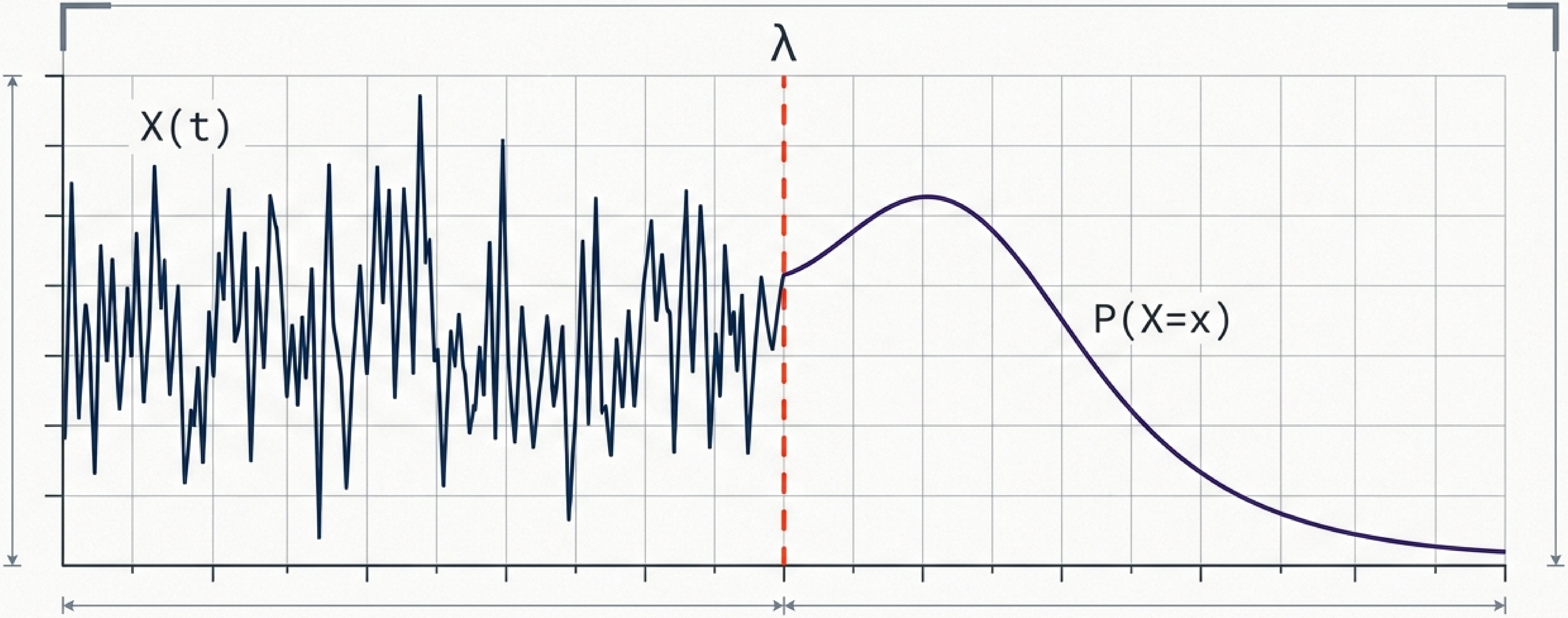
The Blueprint of Uncertainty

Modeling Network Performance with Stochastic Processes

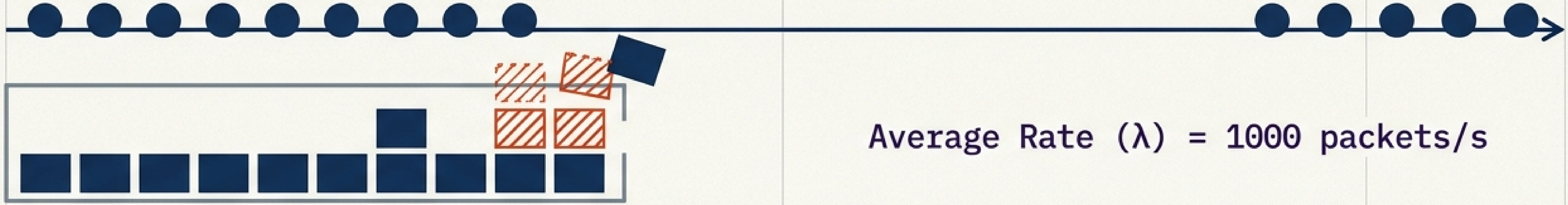


Averages hide the reality of network failure

Network A

Average Rate (λ) = 1000 packets/s

Network B

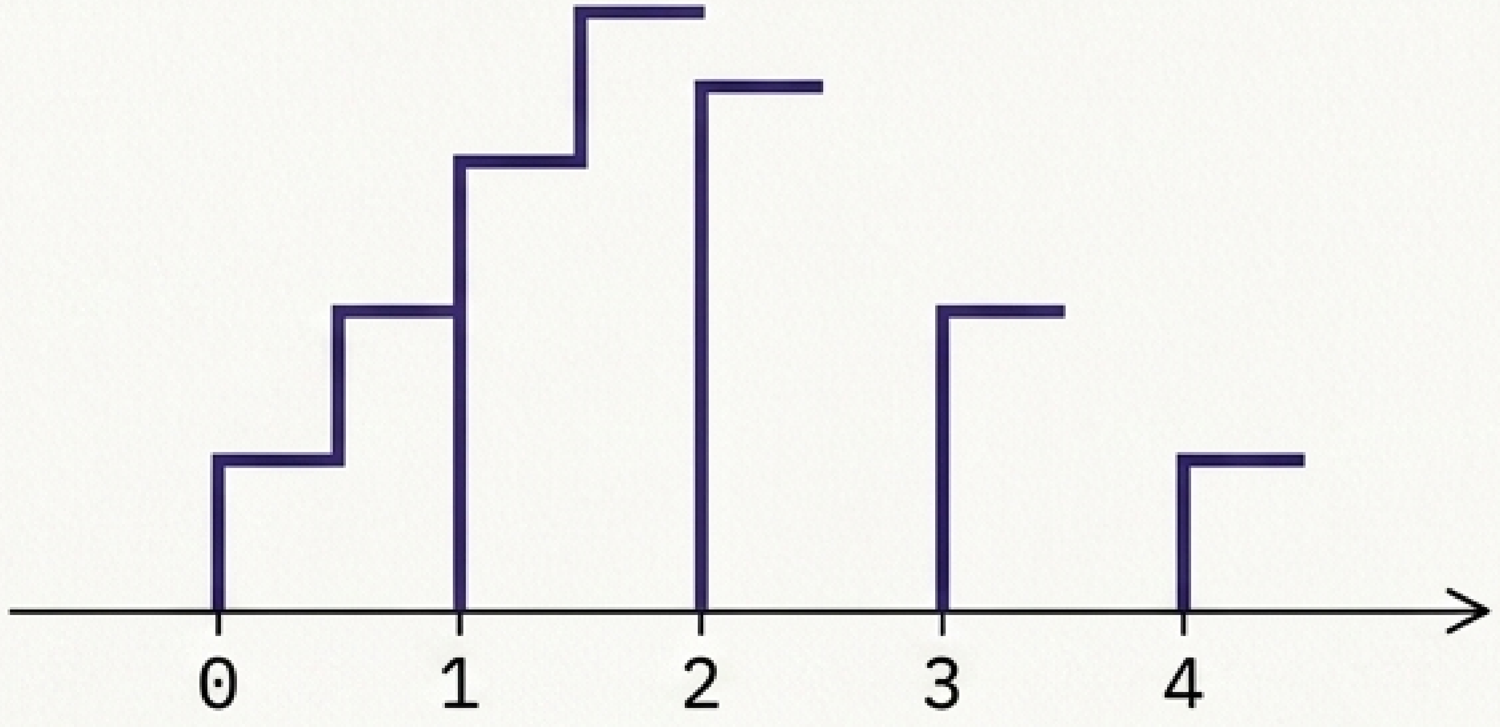


Average Rate (λ) = 1000 packets/s

Traffic bursts instantly spike a queue, creating buffer overflow and packet loss even when the long-term average rate remains perfectly within link capacity.

Capturing uncertainty at a single point in time

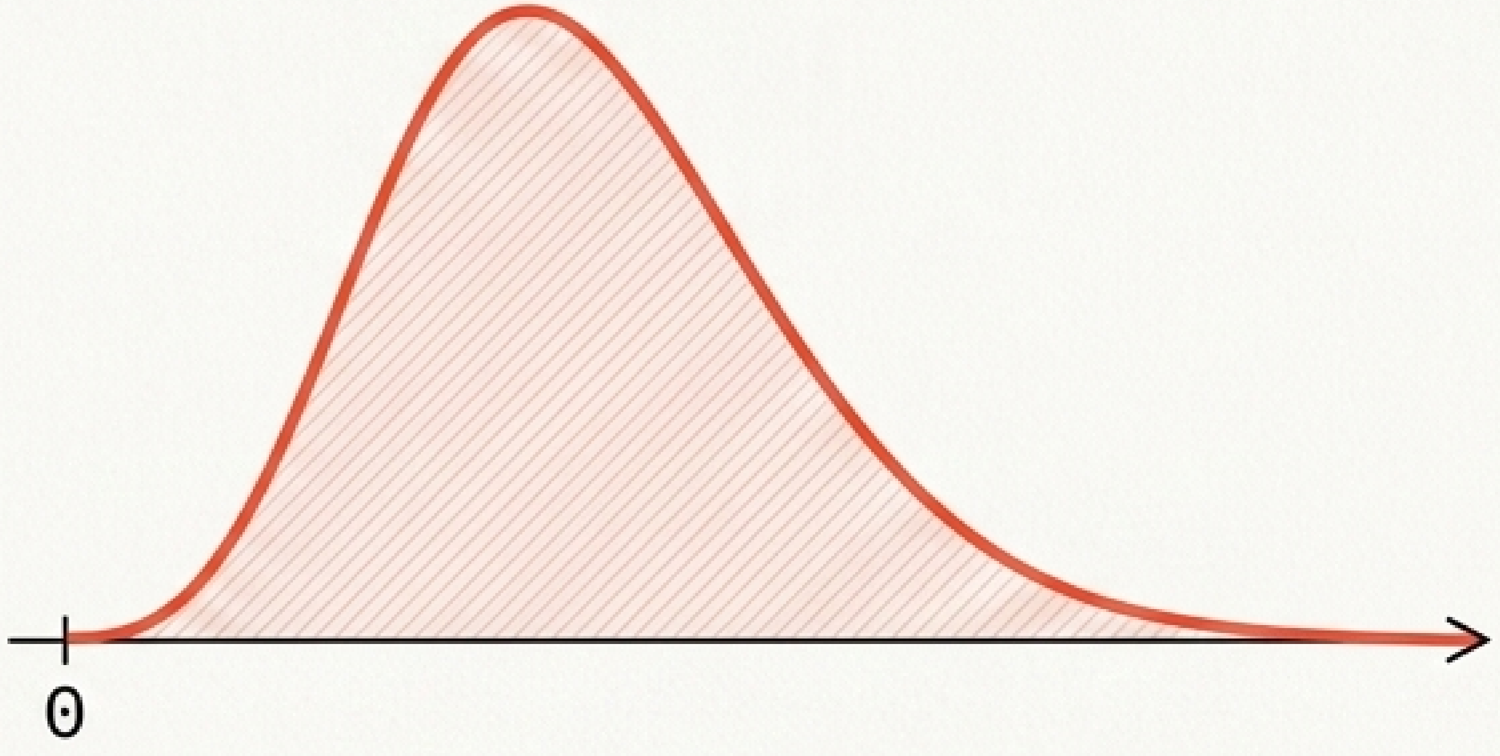
Discrete Random Variables



Example: Number of packets in a router queue.
Values restricted to {0, 1, 2, 3, ...}

$$p_X(x) = P(X=x)$$
$$\sum p_X(x) = 1$$

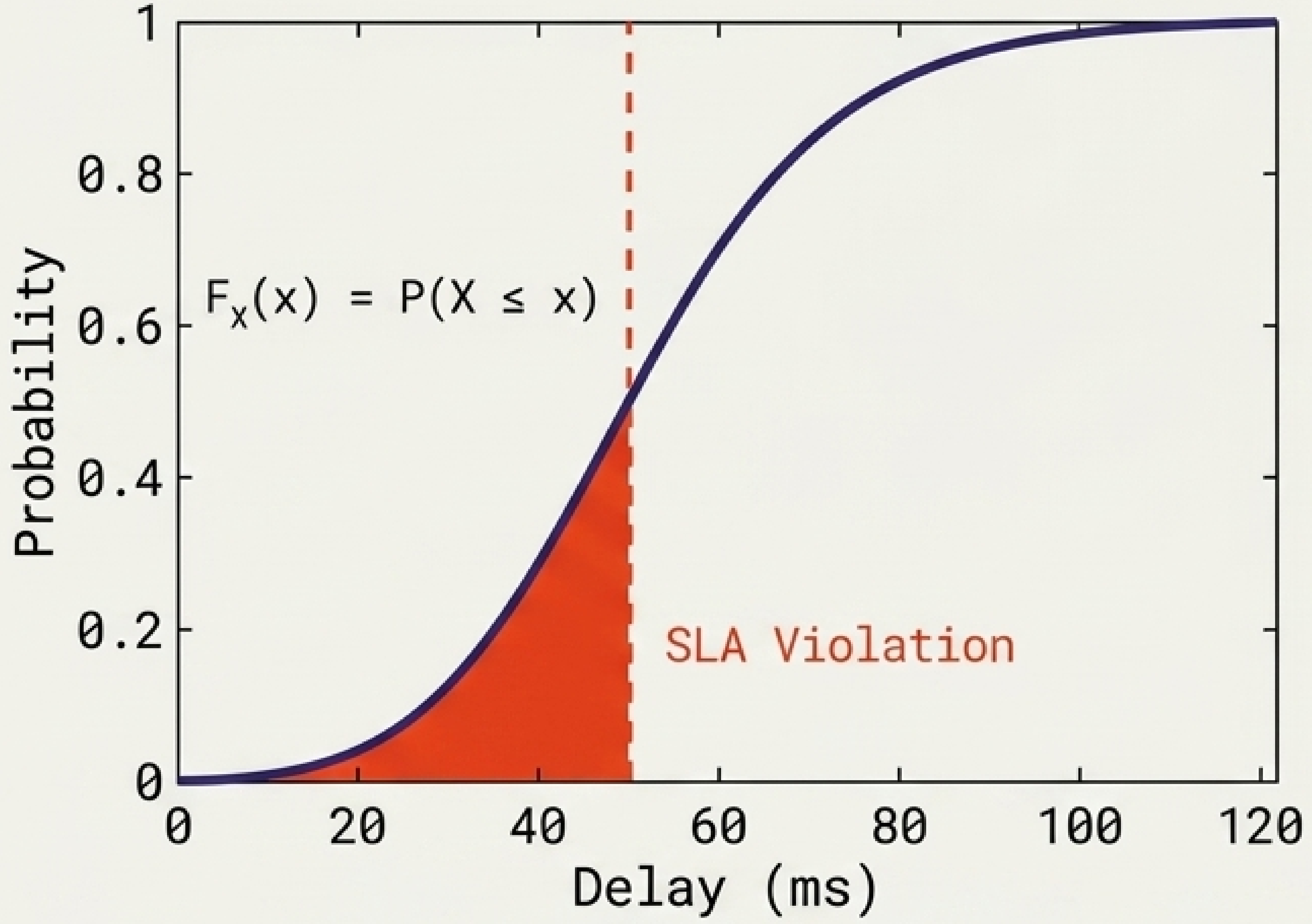
Continuous Random Variables



Example: Packet transmission time.
Values continuous across $[0, \infty)$

$$f_X(x)$$
$$\int f_X(x)\,dx = 1$$

Dispersion and thresholds dictate true performance

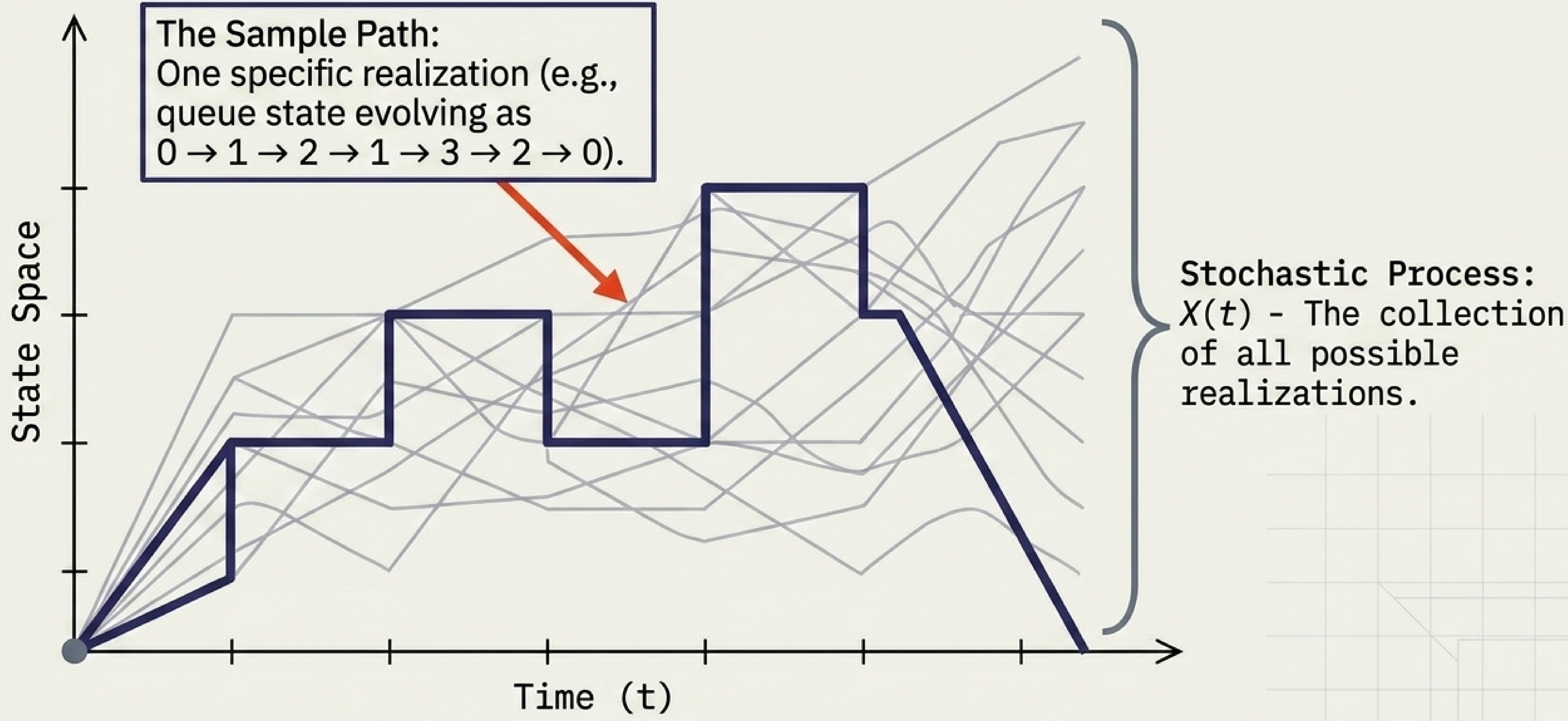


Both systems have identical Expected Delay: $E[D] = 30$ ms	
System A (Low Variance)	$D = \{20, 20, 20, 20\}$ ms
System B (High Variance)	$D = \{5, 5, 5, 65\}$ ms

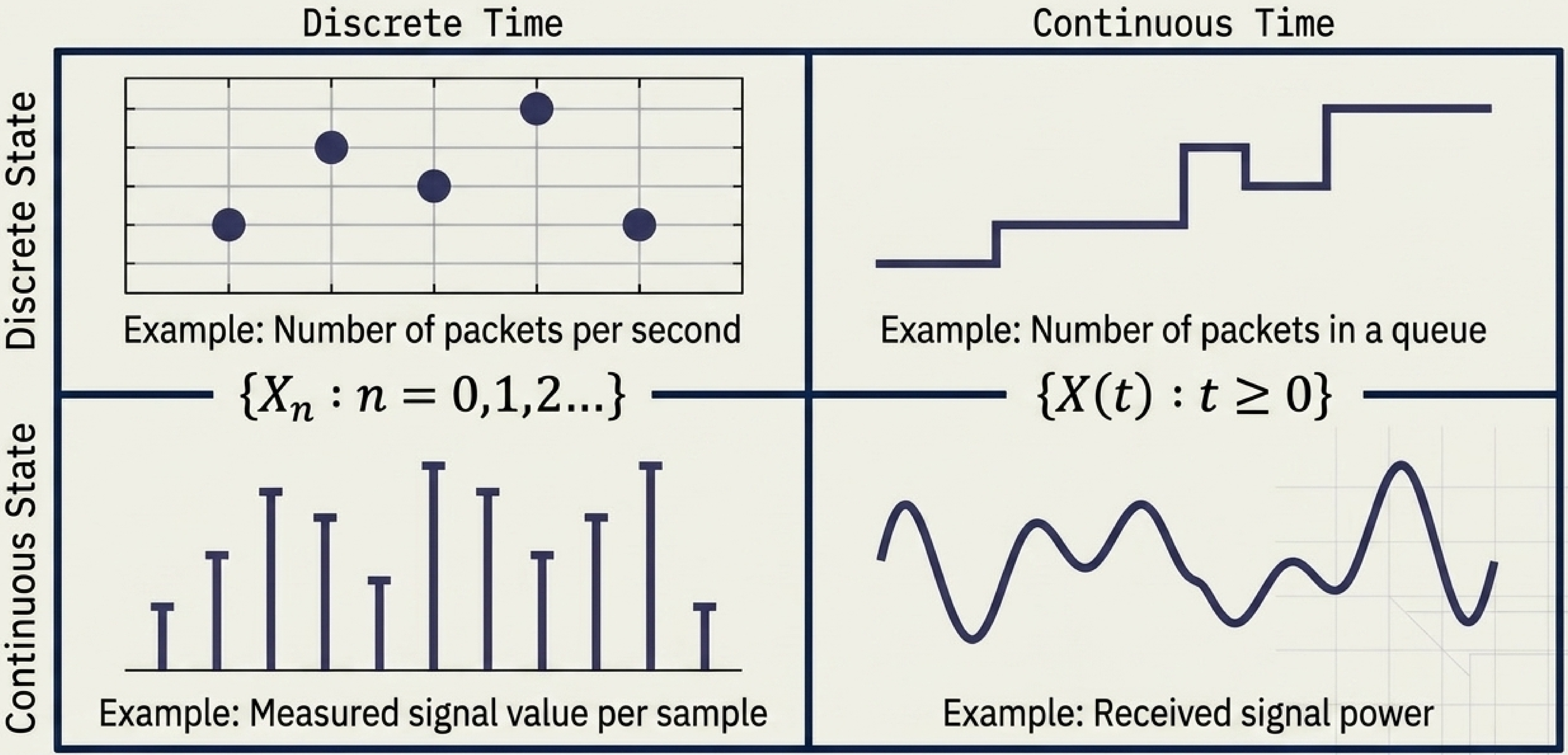
$$\mathrm{Var}(X) = E[(X - E[X])^2]$$

Knowing $P(D \leq 50$ ms$)$ is far more critical for performance analysis than knowing the 30 ms average.

The evolution of randomness over time

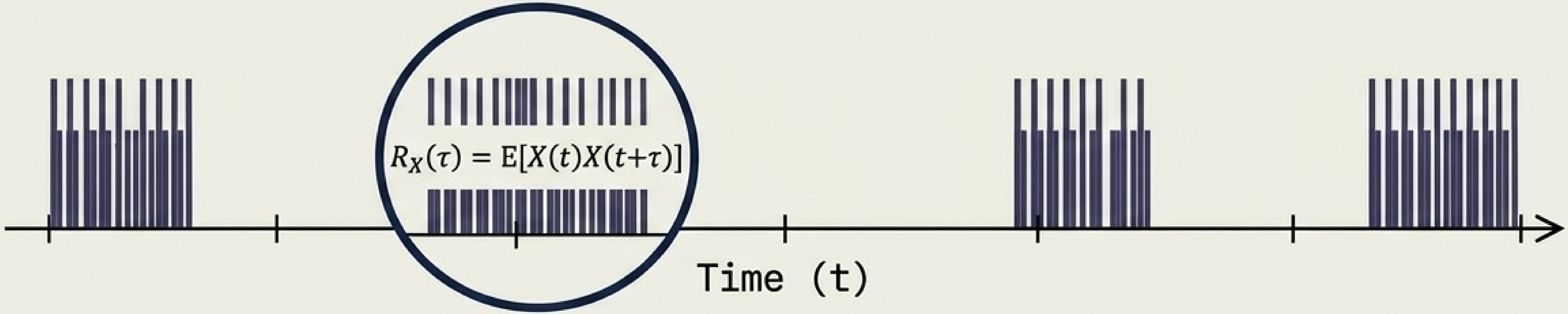


The four domains of network modeling



Gemini Notebook

Autocorrelation creates dangerous traffic bursts



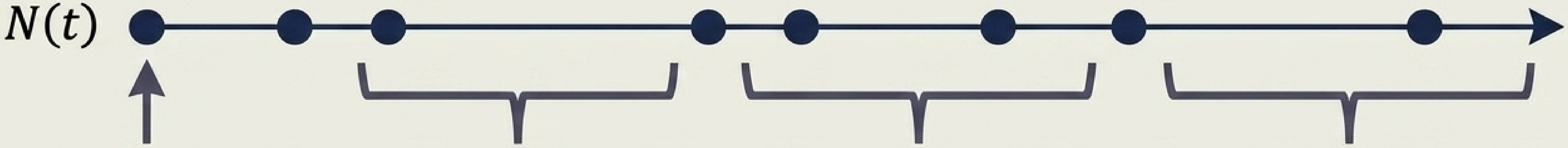
low → high → high → high → high → low

When traffic correlates with itself, values close together in time remain similar.

Burst → Queue Buildup → Buffer Overflow → Packet Loss

Independence is often just a mathematical convenience, not a network reality.

The gold standard for random arrivals



Initial condition:
$N(0) = 0$

Independent increments:
Non-overlapping intervals are entirely independent.

Stationary increments:
Distribution depends only on interval length, not position.

No simultaneous arrivals:
Negligible probability of two events in an infinitesimally small interval.

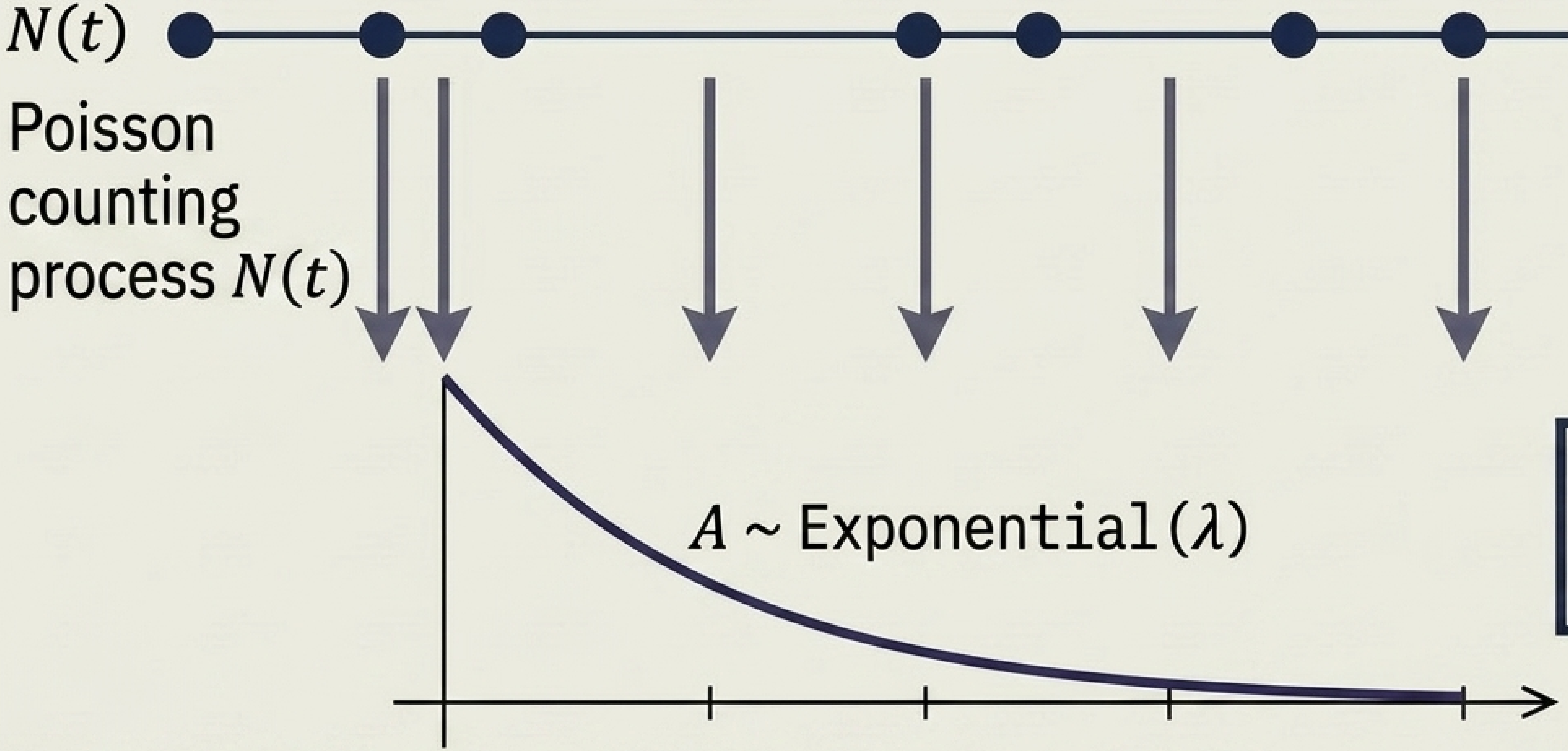
Probability Formula:

$$P(N(t)=n) = \frac{(\lambda t)^n}{n!} * e^{-\lambda t}$$

Mean & Variance Equality:

$$E[N(t)] = Var[N(t)] = \lambda t$$

The duality of arrivals and the time between them

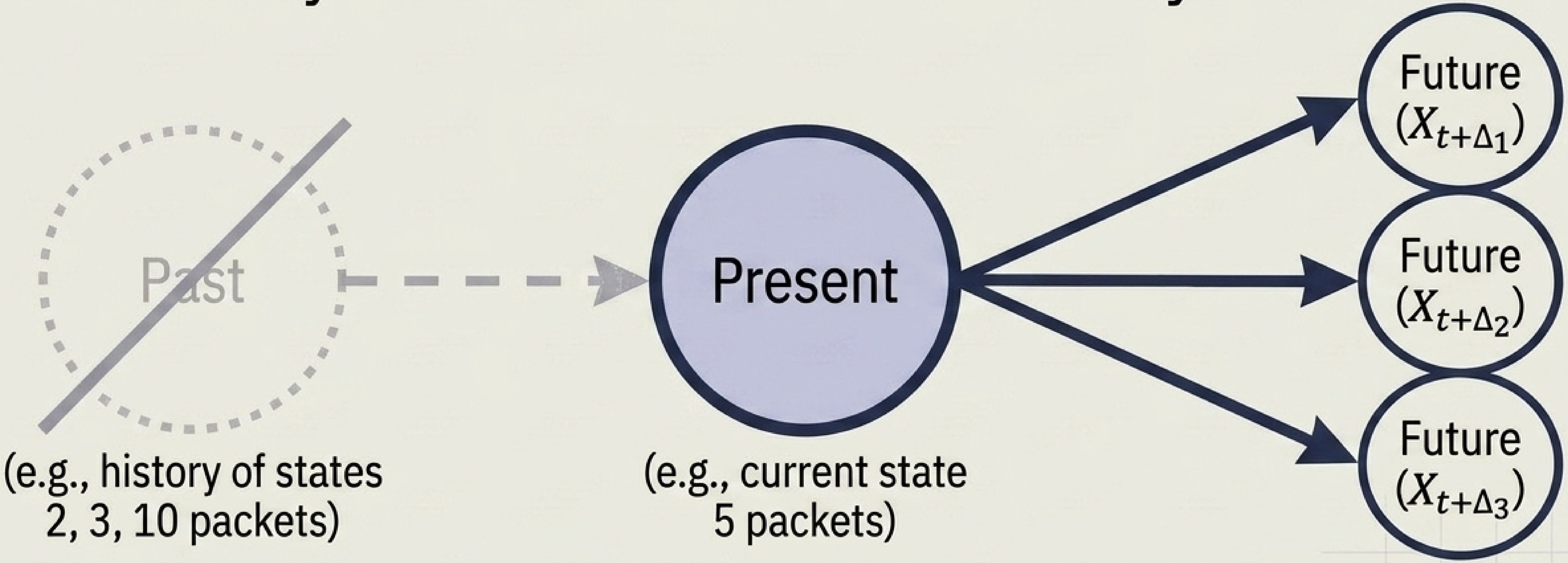


$f_A(a) = \lambda e^{-\lambda a}$

$\mathrm{E}[A] = 1/\lambda$

Poisson arrivals \Longmatrican ⇔ Exponential inter-arrival times.
This relationship forms the foundation of classical queueing theory.

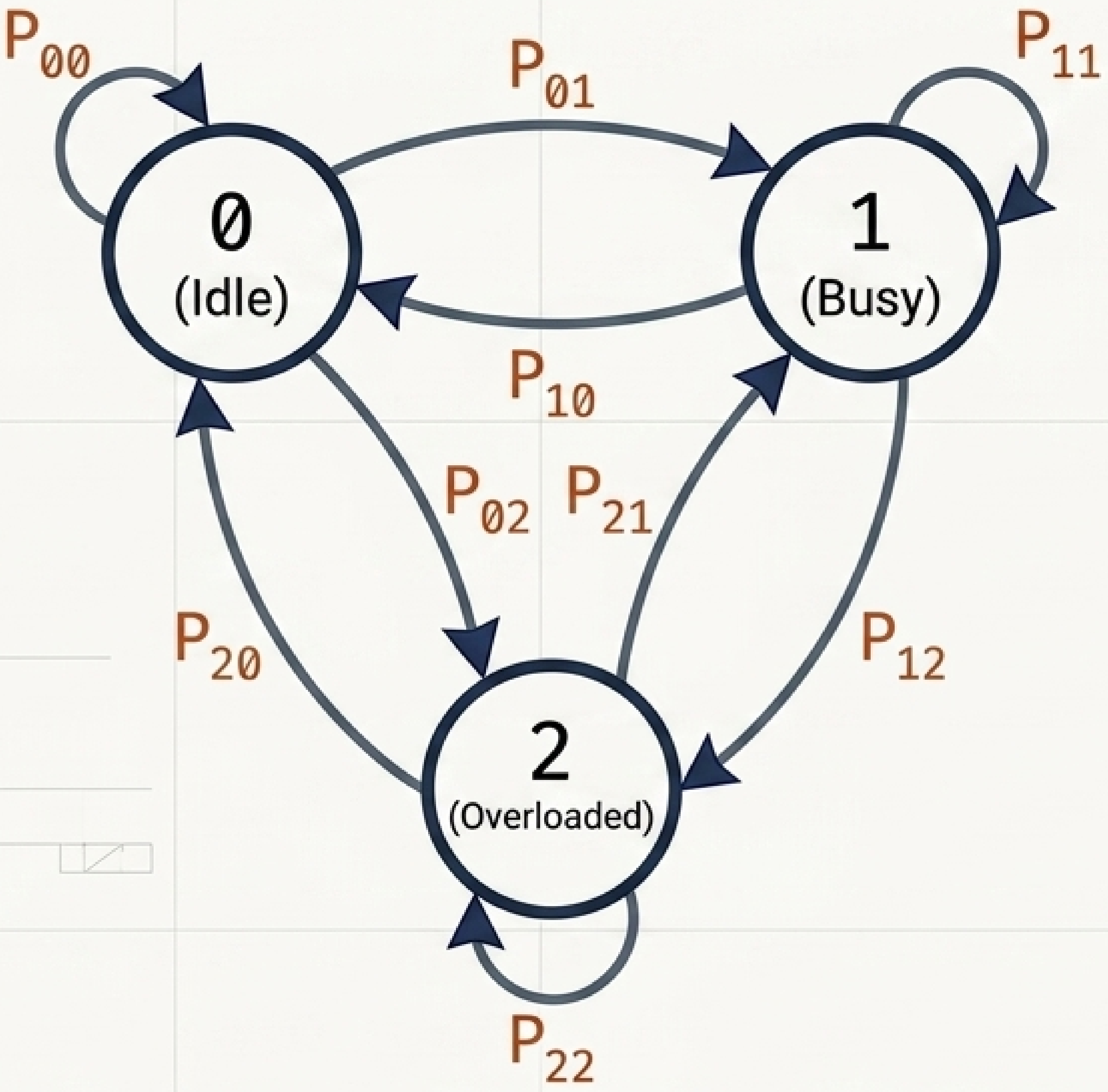
The memoryless nature of Markov systems



$$P(X_{t+\Delta}=x \mid X_t, \text{past history}) = P(X_{t+\Delta}=x \mid X_t)$$

The probability that we must wait an additional t units of time does not depend on how long we have already waited. History is erased.

Mapping discrete network states



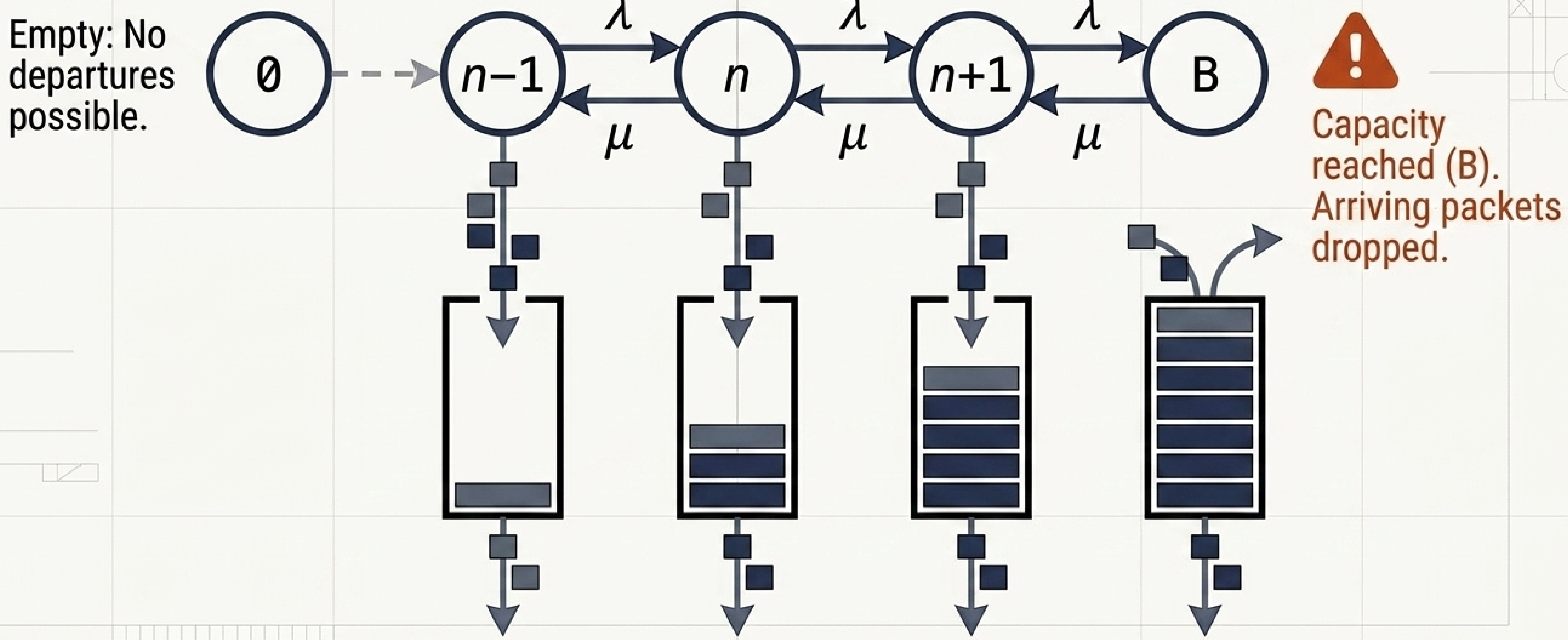
$$\begin{bmatrix} P_{00} & P_{01} & P_{02} \\ P_{10} & P_{11} & P_{12} \\ P_{20} & P_{21} & P_{22} \end{bmatrix}$$

$$P_{ij} = P(X_{n+1}=j \mid X_n=i)$$

(Probability of moving from state i to state j)

Every row must sum to 1: $\sum P_{ij} = 1$

The continuous-time queueing engine



Two paths to truth: Analysis vs. Simulation

Analytical Approach

$$E[X] = \sum x\, P(X = x)$$

$$P(X = n) = \frac{\lambda^n}{n!} e^{-\lambda}$$

$$E[D] = \frac{1}{\mu - \lambda}$$

Calculates exact theoretical quantities using mathematical derivation.

Pros/Cons: Computationally instant, but requires strict mathematical assumptions (like identical distribution).

Simulation Approach

$D_1 = 12\text{ms}$
$D_2 = 45\text{ms}$
$D_3 = 8\text{ms}$
...
$D_N = 14\text{ms}$

Estimates empirical averages by generating random events.

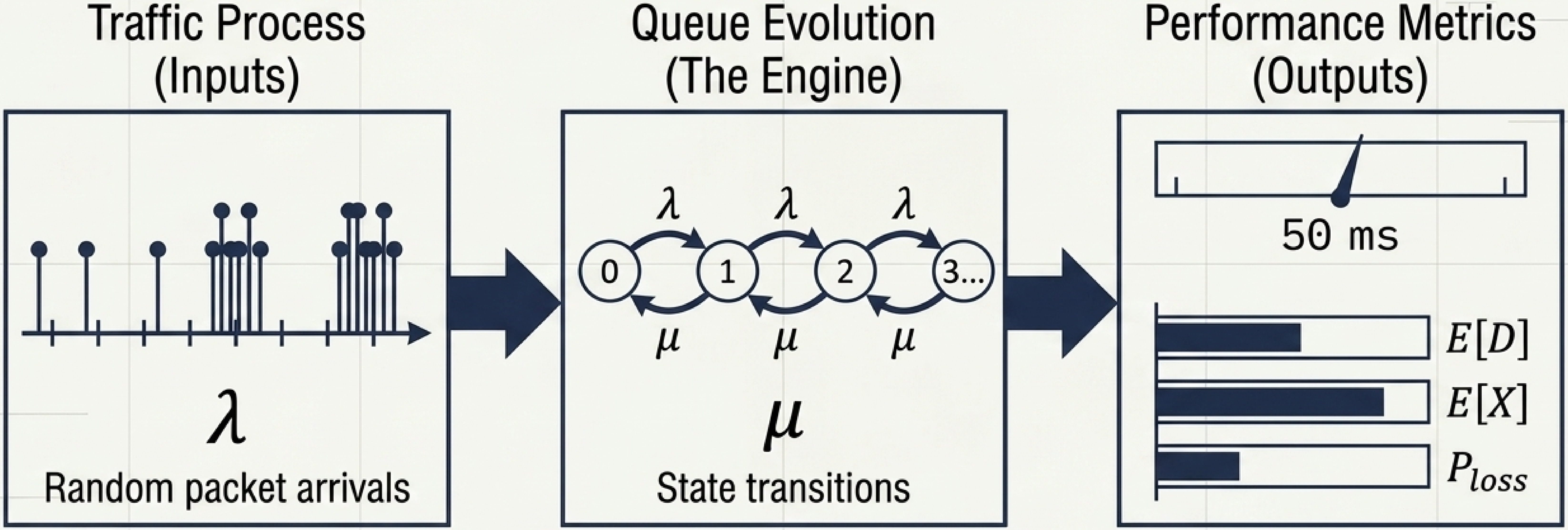
$$\widehat{D} = (1/N) \sum D_i$$

Pros/Cons: Handles immense network complexity, but requires heavy compute time to approach ground truth.

Critical traps in performance modeling

❌	DON'T confuse a **variable** with a **process.** A **variable** is a single random quantity; a **process** is its evolution over time.
❌	DON'T assume **Poisson** blindly. It is mathematically convenient but often fails to capture **heavy-tails** and real **network bursts**.
❌	DON'T ignore **dependence.** Assuming **independence** artificially smooths traffic and massively underestimates actual **queueing delays**.
❌	DON'T confuse **stationarity** with **independence.** A process can be **statistically invariant** over time without its internal **variables** being **independent** of one another.
❌	DON'T focus only on the **mean.** **Average rates** alone cannot predict **queueing behavior**.

From random chaos to predictable metrics



Same mean traffic ≠ Same network performance.